THE ERASURE ENTROPY OF A FORCED COORDINATE UNDER EVERY BIAS

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Abstract. A monotone Boolean function $F_{k,t}$, built from a majority function on k inputs with one hidden coordinate x by substituting threshold functions of $2t$ fresh variables, has x as the only variable lying both in a shortest minterm and in a shortest maxterm. The ratio E_x/Inf_x of the erasure entropy to the influence at x was known to be unbounded when k and t grow together, and to grow linearly in k for $t = 1$. We show that for every fixed $t \geq 1$ it equals $(1 + o(1)) \log_2(1/\mathrm{Inf}_x) = (1 + o(1))(k - 1) \log_2(1/P_t)$, where P_t is the probability that a binomial variable with parameters $2t$ and $\frac{1}{2}$ is at least t. More generally the same holds for the underlying majority function under every rational product bias $P \in (\frac{1}{2}, 1)$, the phase of the relevant coefficients advancing by $\arccos \frac{2P-1}{2P}$, an irrational multiple of π. At $P = \frac{1}{2}$ that angle is $\frac{\pi}{2}$, and the behaviour changes: there the coefficients are values of Krawtchouk polynomials at the middle of their range, half of them vanish, and the ratio is $\frac{1}{2} \log_2(1/\mathrm{Inf}_x) - \frac{1}{2} \log_2 \log_2(1/\mathrm{Inf}_x) + c + O(1/\log_2(1/\mathrm{Inf}_x))$ with $c = \frac{1}{2} + \frac{3}{4} \log_2 e - \frac{1}{2} \log_2 \frac{\pi}{2}$, which we derive from an exact single-sum formula.

1. Introduction

For $f\colon \{0,1\}^n \to \{-1,1\}$ write $f = \sum_S \hat{f}(S)\chi_S$ and let U be a random subset with $\Pr[U = S] = \hat{f}(S)^2$. The Fourier entropy-influence conjecture of Friedgut and Kalai [3] asks for an absolute C with $H(U) \leq C\,\mathbb{E}|U|$. It is known for symmetric functions and read-once decision trees [12], it is preserved under suitable compositions [11], and weaker forms and bounds with extra factors are in [1, 15, 7, 4]; lower bounds on the best constant are in [6]. The erasure entropy of f at a coordinate x is $E_x(f) = H(\mathbf{1}[x \in U] \mid U \setminus \{x\})$, the conditional entropy of one coordinate given the others in the sense of Verdú and Weissman [14], and with $\mathrm{Inf}_x(f) = \Pr[x \in U]$ the conjecture holds if and only if $\sup_f \min_{x:\mathrm{Inf}_x(f)>0} E_x(f)/\mathrm{Inf}_x(f)$ is finite [18]. Since conditioning does not increase entropy,

$$\frac{E_x(f)}{\mathrm{Inf}_x(f)} \leq \frac{h_2(\mathrm{Inf}_x(f))}{\mathrm{Inf}_x(f)} \leq \log_2 \frac{e}{\mathrm{Inf}_x(f)} \tag{1}$$

for every f and x with $\mathrm{Inf}_x(f) > 0$.

The question is which coordinate to take. In [19] several selection rules are shown to fail on monotone functions, among them the rule that takes, among the variables lying both in a shortest minterm and in a shortest maxterm, one of smallest ratio. Fix an odd $k = 2m + 1$, let $y^0 \in \{0,1\}^k$ have $y^0_j = 1$ for $j \leq m$ and $y^0_j = 0$ for $j > m$, let $y^1 = y^0 + e_{m+1}$, let M be MAJ_k with the point y^1 removed from its true set, and

$$\tilde{f}_k(y, x) = M(y) \vee \big(x \wedge [y \geq y^0]\big).$$

For $t \geq 1$ the function $F_{k,t}$ of [19, Section 5] replaces y_j by the threshold function "at least t of $2t$" of a block of fresh variables for $j \leq m$, and by "at least $t + 1$ of $2t$" for $j > m$. It is monotone, and x is the only variable lying both in a shortest minterm and in a shortest maxterm. Moreover [19, proof of Proposition 12] $E_x(F_{k,t})$ and $\mathrm{Inf}_x(F_{k,t})$ equal the erasure entropy and the influence of x in the expansion of $\tilde{f}_k$ in the μ-biased Fourier basis [10, Chapter 8] of the product

measure with $\Pr[y_j = 1] = P_t$ for $j \le m$ and $1 - P_t$ for $j > m$, where

$$P_t = \Pr\big[\mathrm{Bin}(2t, \tfrac12) \ge t\big] = \frac12 + \binom{2t}{t} 2^{-2t-1};$$

the biased form of the conjecture is in [8]. For $t = 1$, $P_1 = \frac34$ and the ratio at x is $(1 + o(1))(k - 1)\log_2 \frac43$ [20]; for fixed $t \ge 2$ it was not known to be unbounded [19, Section 5].

For $P \in (0, 1)$ write $E_x^P(\tilde f_k)$ and $\mathrm{Inf}_x^P(\tilde f_k)$ for the erasure entropy and the influence of x for the law $c(S)^2$, where $c(S)$ are the coefficients of $\tilde f_k$, in ±1 values, in the orthonormal basis of the product measure with $\Pr[y_j = 1] = P$ for $j \le m$, $\Pr[y_j = 1] = 1 - P$ for $j > m$, and x uniform. At $P = \frac12$ these are $E_x(\tilde f_k)$ and $\mathrm{Inf}_x(\tilde f_k)$. Let $\alpha_P = \arccos \frac{2P-1}{2P}$.

Theorem 1. *Let $P \in (\frac12, 1)$ with α_P/π irrational. Then $\mathrm{Inf}_x^P(\tilde f_k) = P^{k-1}$ and, as $k \to \infty$ through odd values,*

$$\frac{E_x^P(\tilde f_k)}{\mathrm{Inf}_x^P(\tilde f_k)} = (1 + o(1))\log_2 \frac{1}{\mathrm{Inf}_x^P(\tilde f_k)} = (1 + o(1))\,(k - 1)\log_2 \frac{1}{P}.$$

The condition holds for every rational $P \in (\frac12, 1)$.

Corollary 2. *For every fixed $t \ge 1$, $\mathrm{Inf}_x(F_{k,t}) = P_t^{k-1}$ and $E_x(F_{k,t})/\mathrm{Inf}_x(F_{k,t}) = (1 + o(1))(k - 1)\log_2(1/P_t)$ as $k \to \infty$. In particular the minterm-maxterm rule of* [19] *has unbounded ratio on the functions $F_{k,t}$ with t fixed.*

The corollary is Theorem 1 with $P = P_t$, which is rational, together with the reduction quoted above. The case $P = \frac34$ is the theorem of [20], and the proof below follows that paper; what makes it work for every P are three identities on the unit circle, (5) below.

At $P = \frac12$ the rotation angle $\alpha_{1/2}$ is $\frac\pi2$ and the argument breaks down for a reason that can be made exact. Write $\Psi(r) = r^2 \log_2(1 + r^{-2}) + \log_2(1 + r^2)$ for $r > 0$ and $\Psi(0) = 0$.

Theorem 3. *Let $k = 2m + 1$. Then $\mathrm{Inf}_x(\tilde f_k) = 2^{-2m}$ and*

$$\frac{E_x(\tilde f_k)}{\mathrm{Inf}_x(\tilde f_k)} = 4^{-m} \sum_{b=0}^{m-1} \binom{2m}{2b+1} \Psi\Big(\binom{m}{b}\binom{2m}{m}\Big/\binom{2m}{2b}\Big).$$

Moreover $\big|E_x(\tilde f_k)/\mathrm{Inf}_x(\tilde f_k) - m + \frac12 \log_2 m\big| \le C$ for an absolute constant C, and

$$\frac{E_x(\tilde f_k)}{\mathrm{Inf}_x(\tilde f_k)} = m - \frac12 \log_2 m + c_0 + O\Big(\frac1m\Big), \qquad c_0 = \frac34 \log_2 e - \frac12 \log_2 \frac\pi2 = 0.75627\ldots,$$

so that, with $L = \log_2(1/\mathrm{Inf}_x(\tilde f_k)) = 2m$, the ratio is $\frac12 L - \frac12 \log_2 L + c_0 + \frac12 + O(1/L)$.

In [19, Proposition 11] the ratio of $\tilde f_k$ was shown to be at least $(\frac18 - o(1))(k - 1)$; the theorem gives the constant $\frac12$. The factor $\frac12$ has a direct meaning: on exactly half of the spectral weight of the derivative in x the average over x has a vanishing coefficient, so there $x \in U$ is determined by $U \setminus \{x\}$ and contributes no entropy. As $t \to \infty$ with k fixed the ratio of $F_{k,t}$ tends to that of $\tilde f_k$ [19, Proposition 12], while for each fixed t the ratio is asymptotic to the bound (1); the two limits do not commute.

2. The ratio as a double sum

All logarithms are to base 2 unless written ln. For $p, q \ge 0$ let $\Phi(p, q) = p \log \frac{p+q}{p} + q \log \frac{p+q}{q}$, with $\Phi(p, 0) = \Phi(0, q) = 0$, so that $\Psi(r) = \Phi(r^2, 1)$. Then $\Psi \ge 0$, $\Psi(r) \ge \log r^2$ for $r \ge 1$, and $\Psi(r) \le \log r^2 + 1 + \log e$ for $r \ge 1$.

Fix $P \in [\frac{1}{2}, 1)$ and put $q = 1 - P$ and $\sigma = \sqrt{Pq}$. In the basis $\prod_{j\in S} \phi_j(y_j)$, with $\phi_j(y) = (y - \mu_j)/\sigma$, $\mu_j = P$ for $j \le m$ and $\mu_j = q$ for $j > m$, and with $+1$ for true, $\tilde{f}_k = \bar{A} + (2x - 1)D$, where

$$D = \mathbf{1}[y = y^0] + \mathbf{1}[y = y^1], \qquad \bar{A} = 2N - 1 + \mathbf{1}[y = y^0] - \mathbf{1}[y = y^1], \qquad N = \mathbf{1}[|y| \ge m + 1].$$

For $S \subseteq [k]$ let $u = |S \cap \{1, \dots, m\}|$ and $v = |S \cap \{m + 2, \dots, k\}|$, and put

$$X(z) = q + Pz, \qquad Y(z) = P + qz,$$
$$H_{uv}(z) = (z - 1)^{u+v} X(z)^{m-u} Y(z)^{m+1-v}, \qquad t_{uv} = \sum_{c \ge m+1} [z^c] H_{uv}(z).$$

Lemma 4. *If $m + 1 \in S$ then $\hat{D}(S) = 0$. If $m + 1 \notin S$ then $\hat{D}(S) = (-1)^v \sigma^{u+v} P^{2m-u-v}$ and $\hat{\bar{A}}(S) = 2\sigma^{u+v} t_{uv} - \mathbf{1}[S = \emptyset] + (P - q)\hat{D}(S)$.*

Proof. This is [20, Lemma 1] with P in place of $\frac{3}{4}$. For a point w, $\widehat{\mathbf{1}[y = w]}(S) = \Pr[y = w] \prod_{j\in S} \phi_j(w_j)$, and $\Pr[y = y^0] = P^{2m+1}$, $\Pr[y = y^1] = P^{2m} q$. If $m + 1 \in S$ the two terms of $\hat{D}(S)$ are $P^{2m+1}\phi_{m+1}(0) = -P^{2m}\sigma$ and $P^{2m} q\, \phi_{m+1}(1) = P^{2m}\sigma$ times the same product, and cancel. Otherwise both points give the product $(-1)^v (q/P)^{(u+v)/2}$, since $\phi_j(1) = \sqrt{q/P}$ for $j \le m$ and $\phi_j(0) = -\sqrt{q/P}$ for $j \ge m + 2$; this gives $\hat{D}(S)$, and the difference of the two indicators has coefficient $(P - q)\hat{D}(S)$. Finally $\mathbb{E}[z^{y_j}\phi_j(y_j)] = \sigma(z - 1)$ and $\mathbb{E}[z^{y_j}] = 1 - \mu_j + \mu_j z$, so $\mathbb{E}[z^{|y|} \prod_{j\in S} \phi_j(y_j)] = \sigma^{u+v} H_{uv}(z)$ when $m + 1 \notin S$, and $\hat{N}(S)$ is the sum of its coefficients at z^c, $c \ge m + 1$. □

Let $\beta_u = \binom{m}{u} q^u P^{m-u}$ and, for $0 \le u, v \le m$,

$$r_{uv} = \frac{\left|2t_{uv} - \mathbf{1}[u = v = 0] + (P - q)(-1)^v P^{2m-u-v}\right|}{P^{2m-u-v}}.$$

Lemma 5. $\mathrm{Inf}_x^P(\tilde{f}_k) = P^{2m}$ *and* $E_x^P(\tilde{f}_k)/\mathrm{Inf}_x^P(\tilde{f}_k) = \sum_{u,v=0}^m \beta_u \beta_v\, \Psi(r_{uv})$.

Proof. As in [20, Lemma 2]: $E_x = \sum_S \Phi(\hat{\bar{A}}(S)^2, \hat{D}(S)^2)$ and $\mathrm{Inf}_x = \sum_S \hat{D}(S)^2$, the terms with $m + 1 \in S$ vanish, there are $\binom{m}{u}\binom{m}{v}$ sets with given (u, v) and $m + 1 \notin S$, $\sum_{u,v} \binom{m}{u}\binom{m}{v}(Pq)^{u+v} P^{2(2m-u-v)} = (Pq + P^2)^{2m} = P^{2m}$, and $\Phi(\hat{\bar{A}}^2, \hat{D}^2) = \hat{D}^2 \Psi(|\hat{\bar{A}}|/|\hat{D}|)$ with $|\hat{\bar{A}}(S)|/|\hat{D}(S)| = r_{uv}$. □

If $u + v \ge 1$ then $Q_{uv} = H_{uv}/(z - 1)$ is a polynomial and $t_{uv} = [z^m]Q_{uv}$: writing $Q_{uv} = \sum_c g_c z^c$, the sum $\sum_{c \ge m+1}(g_{c-1} - g_c)$ telescopes to g_m [20, Lemma 3].

3. The unbiased case

Here $P = q = \frac{1}{2}$, so $X = Y = (1 + z)/2$, the term $(P - q)\hat{D}$ is absent, and $\beta_u = \binom{m}{u} 2^{-m}$. For $0 \le a \le 2m$ let

$$\kappa_a = [z^m](1 - z)^a (1 + z)^{2m-a},$$

the value at a of the Krawtchouk polynomial of degree m for length $2m$ [9, Chapter 5, §7].

Lemma 6. $\kappa_a = 0$ *for odd a, and* $\kappa_{2b} = (-1)^b \binom{m}{b}\binom{2m}{m} / \binom{2m}{2b}$.

Proof. For a variable s,

$$\sum_{a=0}^{2m} \binom{2m}{a} \kappa_a s^a = [z^m]\big((1 + z) + s(1 - z)\big)^{2m} = [z^m]\big((1 + s) + (1 - s)z\big)^{2m} = \binom{2m}{m}(1 - s^2)^m.$$

Comparing coefficients of s^a gives the claim. □

Proof of Theorem 3. By Lemma 5, $\mathrm{Inf}_x = 2^{-2m}$. For $u+v \geq 1$ put $a = u+v-1$; then

$$\frac{2t_{uv}}{P^{2m-u-v}} = 2^{2m-u-v+1}\,[z^m](z-1)^a\Big(\frac{1+z}{2}\Big)^{2m-a} = (-1)^a\kappa_a,$$

so $r_{uv} = |\kappa_{u+v-1}|$. For $u = v = 0$, $t_{00} = \Pr[|y| \geq m+1] = \frac{1}{2}$ and $r_{00} = 0$. The weight of the pairs with $u+v = s$ is $\sum_{u+v=s}\beta_u\beta_v = \binom{2m}{s}4^{-m}$, and by Lemma 6 only odd $s = 2b+1$ give $r \neq 0$, with $r = \binom{m}{b}\binom{2m}{m}/\binom{2m}{2b}$. This is the formula. Since $\binom{2m}{2b} \leq \binom{2m}{m}$, these values satisfy $r \geq \binom{m}{b} \geq 1$.

For the asymptotics write $w_b = \binom{2m}{2b+1}4^{-m}$ and $\delta_b = (2b-m)^2/m$. The odd binomial coefficients of order $2m$ sum to 2^{2m-1}, so $\sum_b w_b = \frac{1}{2}$, and if s is binomial with parameters $2m$ and $\frac{1}{2}$ then $\sum_b w_b\delta_b \leq \mathbb{E}[(s-1-m)^2]/m = \frac{1}{2} + \frac{1}{m}$. We use $h(p) \geq 4p(1-p) = 1-(2p-1)^2$ for the binary entropy h and, for $1 \leq j \leq n-1$,

$$\frac{2^{nh(j/n)}}{\sqrt{8j(n-j)/n}} \leq \binom{n}{j} \leq \frac{2^{nh(j/n)}}{\sqrt{2\pi j(n-j)/n}} \tag{2}$$

[9, Chapter 10, Lemma 7], together with $\binom{2m}{m} \leq 4^m/\sqrt{\pi m}$.

Lower bound. By (2) with $n = m$ and $8b(m-b)/m \leq 2m$, $\log\binom{m}{b} \geq m\,h(b/m) - \frac{1}{2}\log(2m) \geq m - \delta_b - \frac{1}{2}\log m - \frac{1}{2}$ for $1 \leq b \leq m-1$, and this also holds for $b = 0$. As $\Psi(r) \geq \log r^2$,

$$\frac{E_x}{\mathrm{Inf}_x} \geq 2\sum_b w_b \log\binom{m}{b} \geq m - \tfrac{1}{2}\log m - \tfrac{3}{2} - \tfrac{2}{m}.$$

Upper bound. For $1 \leq b \leq m-1$, (2) applied to $\binom{m}{b}$ from above and to $\binom{2m}{2b}$ from below gives

$$\begin{aligned}\log r = \log\binom{m}{b} + \log\binom{2m}{m} - \log\binom{2m}{2b} &\leq 2m - m\,h(b/m) - \tfrac{1}{2}\log m + \tfrac{3}{2} - \log\pi \\ &\leq m + \delta_b - \tfrac{1}{2}\log m + 1,\end{aligned}$$

the factors $b(m-b)/m$ cancelling. For $b = 0$, $r = \binom{2m}{m} \leq 4^m$ and $w_0 = 2m4^{-m}$. With $\Psi(r) \leq \log r^2 + 1 + \log e$,

$$\frac{E_x}{\mathrm{Inf}_x} \leq \sum_{b\geq 1} w_b\big(2m + 2\delta_b - \log m + 3 + \log e\big) + w_0(4m+3) \leq m - \tfrac{1}{2}\log m + C.$$

Finally $\log_2(1/\mathrm{Inf}_x) = 2m$, and $\frac{1}{2}\log m = \frac{1}{2}\log(2m) - \frac{1}{2}$.

The constant. Let $m \geq 2$, write $z_b = 2b-m$, so that $\delta_b = z_b^2/m$, and let C denote absolute constants. First, r is large for every b: since $i \mapsto \binom{m}{i}$ is log-concave, each term of $\binom{2m}{2b} = \sum_i \binom{m}{i}\binom{m}{2b-i}$ is at most $\binom{m}{b}^2$, so $\binom{2m}{2b} \leq (m+1)\binom{m}{b}^2$ and, with $\binom{2m}{m} \geq 4^m/(2\sqrt{m})$ and $\binom{m}{b} \leq 2^m$,

$$r = \binom{m}{b}\binom{2m}{m}\Big/\binom{2m}{2b} \geq \frac{\binom{2m}{m}}{(m+1)\binom{m}{b}} \geq \frac{2^m}{2(m+1)\sqrt{m}}.$$

For $r \geq 1$ and $u = r^{-2}$, $\Psi(r) - \log r^2 - \log e = \log e\,\big(\frac{\ln(1+u)}{u} - 1 + \ln(1+u)\big)$ lies between $-\frac{u}{2}\log e$ and $u\log e$. Hence

$$\frac{E_x}{\mathrm{Inf}_x} = \frac{1}{2}\log e + \sum_b w_b \log r^2 + O(m^3 4^{-m}). \tag{3}$$

Next let $B = \{b : |z_b| \leq m/4\}$. For $b \in B$ we use $\ln\binom{n}{j} = n\,h_e(j/n) - \frac{1}{2}\ln\big(2\pi j(n-j)/n\big) + O(1/n)$ for $\frac{3}{8} \leq j/n \leq \frac{5}{8}$, which follows from Robbins' form of Stirling's formula [13], where $h_e(\frac{1}{2}+\zeta) =$

$\ln 2 - 2\zeta^2 + O(\zeta^4)$ is the binary entropy in nats. With $\zeta = z_b/(2m)$ for both $\binom{m}{b}$ and $\binom{2m}{2b}$, and $\ln(1-4\zeta^2) = O(\zeta^2)$,

$$\ln\binom{m}{b} = m\ln 2 - \frac{z_b^2}{2m} - \frac{1}{2}\ln\frac{\pi m}{2} + \varepsilon_1, \qquad \ln\binom{2m}{2b} = 2m\ln 2 - \frac{z_b^2}{m} - \frac{1}{2}\ln(\pi m) + \varepsilon_2,$$

and $\ln\binom{2m}{m} = 2m\ln 2 - \frac{1}{2}\ln(\pi m) + \varepsilon_3$, with $|\varepsilon_i| \le C(z_b^4/m^3 + z_b^2/m^2 + 1/m)$. Therefore, for $b \in B$,

$$\log r^2 = 2m - \log\frac{\pi m}{2} + \delta_b \log e + \varepsilon_b, \qquad |\varepsilon_b| \le C\Big(\frac{z_b^4}{m^3} + \frac{z_b^2}{m^2} + \frac{1}{m}\Big).$$

For $b \notin B$ both $\log r^2$ and $2m - \log\frac{\pi m}{2} + \delta_b \log e$ lie in $[-\log m, 6m]$, because $1 \le r \le \binom{m}{b}\binom{2m}{m} \le 8^m$ and $\delta_b \le m$. If s is binomial with parameters $2m$ and $\frac{1}{2}$, then $w_b = \Pr[s = 2b+1]$ and $z_b = s - 1 - m$, and for $m \ge 4$ Hoeffding's inequality [5, Theorem 2] gives $\sum_{b\notin B} w_b \le \Pr[|s-m| > \frac{m}{4} - 1] \le 2e^{-(m-4)^2/(16m)}$. Moreover $\mathbb{E}[(s-m)^4] = O(m^2)$, so $\sum_b w_b|\varepsilon_b| = O(1/m)$ over $b \in B$. Finally, $\sum_s (-1)^s\binom{2m}{s}p(s) = 0$ for every polynomial p of degree less than $2m$, so the odd and the even values of s carry equal parts of $\mathbb{E}[(s-1-m)^2] = \frac{m}{2} + 1$, and

$$\sum_b w_b\delta_b = \frac{1}{m}\cdot\frac{1}{2}\Big(\frac{m}{2} + 1\Big) = \frac{1}{4} + \frac{1}{2m}.$$

Putting these together with $\sum_b w_b = \frac{1}{2}$,

$$\sum_b w_b \log r^2 = \frac{1}{2}\Big(2m - \log\frac{\pi m}{2}\Big) + \frac{1}{4}\log e + O\Big(\frac{1}{m}\Big),$$

and by (3) $E_x/\mathrm{Inf}_x = m - \frac{1}{2}\log m + \frac{3}{4}\log e - \frac{1}{2}\log\frac{\pi}{2} + O(1/m)$. □

Remark 7. The proof shows where the weight sits: $r = 0$ exactly when $u+v$ is even, which is half of the weight, and on the other half $\log r^2$ is $2m - \log m + O(1)$ for typical b. The constant c_0 collects three contributions: the Gaussian shape of the three binomial coefficients, which gives $-\frac{1}{2}\log\frac{\pi}{2}$; their curvature, which makes $\log r^2$ grow by $\delta_b \log e$ away from the centre and contributes $\frac{1}{4}\log e$; and the gap $\Psi(r) - \log r^2 \to \log e$, which contributes $\frac{1}{2}\log e$.

4. The tail on the unit circle

From now on $P \in (\frac{1}{2}, 1)$. For $z = e^{\mathrm{i}\theta}$ and $w = \cos\theta$,

$$|z-1|^2 = 2 - 2w, \qquad |X(z)|^2 = |Y(z)|^2 = R(w) := P^2 + q^2 + 2Pqw,$$

and $X(z) = z\,\overline{Y(z)}$. Put $n_1 = u + v - 1$ and $n_2 = 2m + 1 - u - v$, so $n_1 + n_2 = 2m$. Then $|Q_{uv}(e^{\mathrm{i}\theta})| = e^{\rho(\cos\theta)}$ with

$$\rho(w) = \frac{n_1}{2}\ln(2-2w) + \frac{n_2}{2}\ln R(w), \qquad \rho''(w) = -\frac{n_1}{2(1-w)^2} - \frac{2P^2q^2n_2}{R(w)^2} \le -2P^2q^2n_2,$$

using $R \le 1$. So ρ is strictly concave on $[-1,1)$, with maximum at

$$w_0 = \frac{2Pq\,n_2 - (P^2+q^2)\,n_1}{4Pq\,m}$$

when this lies in $(-1,1)$. A short computation gives $2 - 2w_0 = n_1/(2Pqm)$ and $R(w_0) = n_2/(2m)$, hence

$$\rho(w_0) = \frac{n_1}{2}\ln\frac{n_1}{2Pqm} + \frac{n_2}{2}\ln\frac{n_2}{2m}. \tag{4}$$

The law β_u has mean mq and variance mPq. For $\lambda \ge 1$ let W_λ be the set of (u,v) with $|u - mq| \le \lambda\sqrt{m}$ and $|v - mq| \le \lambda\sqrt{m}$. Below, c, C are positive constants that may depend on

P and λ but not on m or on $(u,v) \in W_\lambda$, and m is large enough for each statement. On W_λ, $n_1 = 2mq + O(\sqrt{m})$ and $w_0 = w^* + O(m^{-1/2})$, where

$$w^* = \frac{2P-1}{2P} = \cos\alpha_P \in (0, \tfrac{1}{2}).$$

Put $\theta_0 = \arccos w_0$, $z_0 = e^{\mathrm{i}\theta_0}$, $\theta^* = \alpha_P$ and $z^* = e^{\mathrm{i}\theta^*}$. Since $1 - w^* = 1/(2P)$, the real part of $(z^*-1)(q + P\bar{z}^*)$ is $(P-q)(1-w^*) = w^*$ and its imaginary part is $\sin\theta^*$, and $R(w^*) = P^2 + q^2 + q(P-q) = P$; so

$$(z^*-1)\,\overline{X(z^*)} = z^*, \qquad |X(z^*)|^2 = P, \qquad Y(z^*) = z^*\,\overline{X(z^*)}. \tag{5}$$

Lemma 8. *On W_λ, $\rho(w_0) + (n_2 - 1)\ln\frac{1}{P} \geq (m-1)\ln\frac{1}{P}$.*

Proof. Regard the left side, by (4), as a function $G(n_1)$ of a real $n_1 \in (0, 2m)$ with $n_2 = 2m - n_1$. Then $G'(n_1) = \frac{1}{2}\ln\frac{n_1}{Pq\,n_2} + \ln P$, which vanishes at $n_1 = 2mq$, and $G'' = \frac{1}{2n_1} + \frac{1}{2n_2} > 0$. So $G \geq G(2mq) = mq\ln\frac{1}{P} + mP\ln P + (2mP-1)\ln\frac{1}{P} = (m-1)\ln\frac{1}{P}$. □

Fix $(u,v) \in W_\lambda$ and let $\Lambda(\theta) = \ln Q_{uv}(e^{\mathrm{i}\theta}) - \mathrm{i}m\theta$ for complex θ with $|\theta - \theta^*| \leq \delta$, where $\delta > 0$ is so small that $e^{\mathrm{i}\theta}$ stays away from 1, $-q/P$ and $-P/q$, the zeros of $z-1$, X and Y; the logarithm is the branch continuous in this disc. Then $|\Lambda''|, |\Lambda'''| \leq Cm$ on the disc. For real θ, $\operatorname{Re}\Lambda(\theta) = \rho(\cos\theta)$, so $\operatorname{Re}\Lambda'(\theta_0) = 0$ and $\operatorname{Re}\Lambda''(\theta_0) = \rho''(w_0)\sin^2\theta_0 \leq -cm$. Put

$$b = \operatorname{Im}\Lambda'(\theta_0), \qquad A = -\Lambda''(\theta_0), \qquad T_{uv} = \frac{1}{\sqrt{2\pi A}}\exp\Big(\Lambda(\theta_0) - \frac{b^2}{2A}\Big).$$

Lemma 9. *On W_λ: $|b| \leq C\sqrt{m}$, $|T_{uv}| \geq c\,m^{-1/2}e^{\rho(w_0)}$, and $t_{uv} = 2\operatorname{Re}T_{uv} + O(|T_{uv}|\,m^{-1/5})$.*

Proof. With $a = u/m$ and $a' = v/m$, $\Lambda'(\theta)/m$ differs by $O(1/m)$ from

$$\mathrm{i}z\Big(\frac{a+a'}{z-1} + \frac{(1-a)P}{X(z)} + \frac{(1-a')q}{Y(z)}\Big) - \mathrm{i}, \qquad z = e^{\mathrm{i}\theta},$$

a smooth function of (a, a', θ). It vanishes at $a = a' = q$, $\theta = \theta^*$: by (5), $1/(z^*-1) = \overline{X^*}/z^*$, $1/X^* = \overline{X^*}/P$ and $1/Y^* = X^*/(Pz^*)$, where $X^* = X(z^*)$, so

$$z^*\Big(\frac{2q}{z^*-1} + \frac{P^2}{X^*} + \frac{Pq}{Y^*}\Big) = 2q\overline{X^*} + Pz^*\overline{X^*} + qX^* = 3q^2 + P^2 + 4Pq\,w^* = 1.$$

The rest is the proof of [20, Lemma 5]. Since $a-q$, $a'-q$ and $\theta_0 - \theta^*$ are $O(m^{-1/2})$, $|\Lambda'(\theta_0)| \leq C\sqrt{m}$, and $\operatorname{Re}(b^2/A) = b^2\operatorname{Re}A/|A|^2 \leq C$, which gives the lower bound on $|T_{uv}|$. By Cauchy's formula [2, Chapter VIII], $t_{uv} = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{\Lambda(\theta)}d\theta$. On $J_+ = \{|\theta - \theta_0| \leq m^{-2/5}\}$, Taylor's formula $\Lambda(\theta_0 + h) = \Lambda(\theta_0) + \mathrm{i}bh - \frac{1}{2}Ah^2 + O(m|h|^3)$ and a Gaussian integral give $T_{uv} + O(|T_{uv}|m^{-1/5})$; the integral over $J_- = -J_+$ is its complex conjugate, as Q_{uv} has real coefficients. Off $J_+ \cup J_-$, $|\cos\theta - w_0| \geq c|\theta \mp \theta_0|$ because θ_0 stays away from 0 and π, and $\rho'' \leq -cm$ gives a modulus at most $e^{\rho(w_0) - cm^{1/5}}$. □

Lemma 10. *If (u,v) and $(u+1,v)$ lie in W_λ, then $\arg T_{u+1,v} - \arg T_{uv} = \alpha_P + O(m^{-1/2})$ modulo 2π.*

Proof. Let $g(z) = (z-1)/X(z)$, so $Q_{u+1,v} = gQ_{uv}$. Passing from (u,v) to $(u+1,v)$ changes w_0 by $-1/(4Pqm)$, so θ_0 moves by $O(1/m)$, and as in [20, Lemma 6] $\arg T$ moves by $\operatorname{Im}\ln g(z_0) + O(m^{-1/2})$. By (5), $g(z^*) = (z^*-1)\overline{X^*}/|X^*|^2 = z^*/P$, so $\arg g(z^*) = \theta^* = \alpha_P$, and $z_0 - z^* = O(m^{-1/2})$. □

If P is rational then so is $\cos\alpha_P = (2P-1)/(2P)$, and it lies in $(0, \frac{1}{2})$; by Niven's theorem [17] a rational multiple of π with rational cosine has cosine in $\{0, \pm\frac{1}{2}, \pm 1\}$, so α_P/π is irrational. This is the last claim of Theorem 1.

5. Proof of Theorem 1

The influence is Lemma 5 and the upper bound is (1). The lower bound is the argument of [20, Section 4], which we repeat with the constants of a general P. Fix $\varepsilon \in (0, \frac{1}{4})$, $\lambda \geq 1$ and $L \geq 1$ so large that for every real ψ_0 at most $3\varepsilon L$ of the numbers $\ell = 0, \dots, L-1$ satisfy $|\cos(\psi_0 + \ell\alpha_P)| < 2\varepsilon$; this is possible because $\ell\alpha_P/\pi$ is equidistributed modulo 1 [16]. Fix v and cut the u with $(u, v) \in W_\lambda$ into blocks of L consecutive values. By Lemma 10, in each block at most $3\varepsilon L$ values have $|\cos \arg T_{uv}| < \varepsilon$ once m is large. On W_λ, $\beta_{u+1}/\beta_u = (m-u)q/((u+1)P) = 1 + O(m^{-1/2})$, so the weights in a block differ by a factor at most 2, and the bad values and the left-over values carry $\beta_u\beta_v$-weight at most $6\varepsilon + o(1)$ in all.

Let $(u, v) \in W_\lambda$ have $|\cos \arg T_{uv}| \geq \varepsilon$. By Lemma 9, $|t_{uv}| \geq \varepsilon |T_{uv}| \geq c\varepsilon m^{-1/2} e^{\rho(w_0)}$, and with $2m - u - v = n_2 - 1$ and Lemma 8,

$$r_{uv} \geq \frac{2|t_{uv}|}{P^{n_2-1}} - (P - q) \geq c\varepsilon m^{-1/2} P^{-(m-1)} - 1,$$

so $\Psi(r_{uv}) \geq \log r_{uv}^2 \geq 2m \log \frac{1}{P} - \log m - C_\varepsilon$. By Chebyshev's inequality W_λ has weight at least $(1 - Pq/\lambda^2)^2$, and since $\Psi \geq 0$, Lemma 5 gives

$$\frac{E_x^P}{\mathrm{Inf}_x^P} \geq \Big(\big(1 - \tfrac{Pq}{\lambda^2}\big)^2 - 6\varepsilon - o(1)\Big)\Big(2m \log \frac{1}{P} - \log m - C_\varepsilon\Big).$$

Dividing by $\log(1/\mathrm{Inf}_x^P) = 2m \log \frac{1}{P}$ and letting $m \to \infty$, then $\lambda \to \infty$ and $\varepsilon \to 0$, gives $\liminf \geq 1$. □

6. Numerical checks

Lemma 5 reduces the ratio to $(m+1)^2$ terms whose ingredients are integers over powers of the denominator of P; the code computes them exactly in integer arithmetic and evaluates each term of Φ as $p \log(1 + q/p)$ with a $\log(1 + \cdot)$ routine. For $t = 1$ the values agree with those of [20], and for $t = 0, 1, 2, 3$ at $k \leq 27$ with a rational-arithmetic implementation. The quotient of $E_x(F_{k,t})/\mathrm{Inf}_x(F_{k,t})$ by $(k-1)\log_2(1/P_t)$, with $P_2 = \frac{11}{16}$, $P_3 = \frac{21}{32}$ and $P_4 = \frac{163}{256}$, is:

k	41	81	121	161	201	251	301
$t = 2$	0.849	0.899	0.922	0.936	0.946	0.954	0.960
$t = 3$	0.854	0.902	0.928	0.941	0.950	0.958	0.964
$t = 4$	0.858	0.909	0.931	0.944	0.953	0.960	0.966

For $\tilde{f}_k$ the formula of Theorem 3 agrees with the double sum at $k = 13$, 41, 81 and 121 to six decimals, and the identity of Lemma 6 was checked against direct expansion for $m \leq 11$. The quotient by $k - 1$ is 0.466, 0.476, 0.482, 0.490, 0.496 and 0.499 at $k = 41$, 81, 121, 301, 1001 and 4001, and $E_x/\mathrm{Inf}_x - m + \frac{1}{2}\log_2 m$ is 0.7796, 0.7677, 0.7639, 0.7593, 0.7572 and 0.7565 at the same values of k. The difference from $c_0 = 0.75627$, multiplied by m, is 0.467, 0.459, 0.456, 0.453, 0.451 and 0.4510 at these k, and 0.4509 at $k = 8001$, in line with the error term $O(1/m)$ of Theorem 3.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [23] and Math Brain [21]. The family $F_{k,t}$ is from [19] and the method for $P = \frac{3}{4}$ from [20]. Claude Opus 5.5 (Anthropic) found Theorems 1 and 3 and their proofs, wrote the numerical checks and prepared the manuscript. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-forced-coordinate-bias`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [21].

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